Cloud Computing Evaluation Paper

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Cloud Computing

Cloud computing can be divided into deployment models and service models depending on how an organization wants to use and manage its resources. The main deployment models are public, private, community, and hybrid cloud. A public cloud uses resources provided by a third-party provider and can be useful when an organization wants flexibility and scalability without maintaining all of its own physical infrastructure. A private cloud is dedicated to one organization and may be more appropriate when greater control, customization, or security is needed. A community cloud is shared by organizations with similar requirements or concerns, while a hybrid cloud combines different cloud environments. A hybrid model can be useful when an organization wants to keep certain information or workloads in a more controlled environment while still using public cloud resources for other needs **(Mell & Grance, 2011).**

Cloud service models describe how much of the technology is managed by the cloud provider. Infrastructure as a Service (IaaS) provides resources such as computing, storage, and networking while allowing the organization to maintain more control over its software and applications. Platform as a Service (PaaS) provides an environment where developers can build and deploy applications without having to manage as much of the underlying infrastructure. Software as a Service (SaaS) provides complete applications that users can access without managing the infrastructure or platform themselves. IaaS is useful when an organization needs more control and flexibility, PaaS is useful when the main goal is developing applications, and SaaS is appropriate when users mainly need access to finished software **(Mell & Grance, 2011).**

Benefits and Drawbacks of the Cloud

Cloud deployment models have several benefits compared with keeping all computing resources on-premises. One of the biggest advantages is scalability because an organization can increase or decrease its computing resources as its needs change. Cloud computing can also reduce the amount of physical hardware an organization needs to purchase and maintain. Resources can be made available more quickly, and employees may have easier access to applications and information from different locations. With an on-premises model, the organization has to purchase, maintain, and eventually replace much of its own hardware, which can require more time and money.

There are also drawbacks to using the cloud. An organization becomes more dependent on its cloud provider and internet connectivity, and outages could temporarily prevent users from accessing important resources. Costs can also become difficult to predict because they depend on how cloud resources are being used. On-premises environments can provide an organization with more direct control over its hardware, data, and configurations. Cloud computing also introduces security and privacy concerns because responsibility for security is divided between the provider and customer. Because of these differences, cloud deployment is not automatically better than an on-premises environment, and the better option depends on the organization's needs.

Cloud Deployment Models

Each cloud deployment model has different benefits and risks. Public clouds can provide scalability and reduce the need for an organization to maintain its own physical infrastructure, but organizations have less direct control over the underlying environment. Private clouds provide greater control and can be designed around the specific security or technical

requirements of one organization, although they can require more resources and expertise to operate. Community clouds allow organizations with similar requirements to share infrastructure and costs, but the participating organizations must agree on how the environment will be managed and secured.

Hybrid clouds combine different cloud environments and can give an organization more flexibility. For example, sensitive information could remain in a private environment while other applications use public cloud resources. The drawback is that managing multiple environments can make the organization's systems more complicated. Data may need to move securely between environments, and administrators have to make sure the different systems work together correctly. The risks and benefits of each deployment model therefore need to be compared with the type of information the organization handles and the amount of control, flexibility, and security it requires.

Considerations of Cloud Computing

An organization should consider both organizational and technical issues before switching to a cloud model. Organizational considerations include the cost of migration, employee training, changes to existing workflows, and deciding who will be responsible for managing cloud resources. Employees may need to learn new systems, and an organization may need new policies for accessing and managing information. The organization should also consider whether the long-term costs of the cloud fit its budget instead of focusing only on the initial cost of migration.

Technical considerations include security, privacy, compatibility, network reliability, and the process of moving existing applications and data. An organization needs to determine whether its current systems will work correctly in the new environment and how data will be transferred without being lost or exposed. Backup and recovery plans are also important in case something goes wrong during or after migration. Security responsibilities must be clearly understood because moving to a cloud provider does not mean that the provider becomes responsible for every part of security. For example, AWS describes cloud security as a shared responsibility between the provider and customer (Amazon Web Services [AWS], n.d.).

Big Data vs. Structured Data

Structured data is generally organized into a consistent format, such as rows and columns in a relational database, while big data can contain much larger amounts of structured, semi-structured, and unstructured information from many different sources. Structured data may come from sources such as transactions or forms that follow predefined fields. Big data can also come from sources such as sensors, logs, websites, social media, and connected devices. Because of its volume and variety, big data may require distributed storage and processing instead of relying only on a traditional relational database. Preprocessing structured data can involve correcting missing values, removing duplicates, checking data types, and preparing columns for analysis. Big data can require similar cleaning steps, but the amount and variety of information can require additional processing to combine and standardize data from different sources.

In uCertify Lab 1.3.2, I created a smaller subset of the diamonds DataFrame. After loading the CSV file into Pandas, I used df.loc[16:20, ['carat', 'cut', 'depth', 'price']] to select rows

16 through 20 and only the carat, cut, depth, and price columns. This produced a smaller portion of the original dataset containing only the information needed for the activity. Separating a large dataset into smaller subsets can make the data easier to work with because unnecessary rows and columns do not need to be processed every time. It can also make it easier to inspect the data, identify problems, and focus an analysis on the variables that are actually relevant.

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Volume, Variety, and Velocity of Big Data

The three Vs of big data are volume, variety, and velocity. Volume describes the amount of data, variety refers to the different forms and sources of data, and velocity describes how quickly data is generated and needs to be processed. As these increase, conventional methods can become less practical. A dataset may become too large to efficiently process on one computer, different types of information may need to be cleaned or converted before they can be analyzed together, and rapidly generated data may need to be processed continuously. Scale can also affect usability because having more data is not necessarily useful if it cannot be stored, organized, processed, and analyzed efficiently.

In uCertify Lab 1.2.2, I loaded the weight dataset into a Pandas DataFrame and grouped the data using group = df.groupby('sex'). I then used group.mean() to calculate the mean weight for each of the three groups, which produced mean weights of approximately 33.58, 31.65, and 37.19. If the same type of calculation needed to be performed for hundreds of variables, it would not make sense to manually calculate each value. Grouping and aggregation functions could perform those calculations across many columns at once, while larger datasets could require tools designed for distributed processing. The three Vs make this more important because

increasing volume creates more information to process, variety creates more types of information to prepare, and velocity can require calculations to happen quickly as new information is collected.

References

Amazon Web Services. (n.d.). *Shared responsibility model.* https://aws.amazon.com/compliance/shared-responsibility-model/

Mell, P., & Grance, T. (2011). *The NIST definition of cloud computing* (NIST Special Publication 800-145). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.SP.800-145